

1 My solutions for exercises

I'm practicing proof writing as well as problem solving here, so the writing style can vary from one solution to another.

1.1 Exercise 2.2.3

1.1.1 (a) $a \geq a$.

Proof. We seek an m such that $a = a + m$. Clearly 0 is such an m , and hence $a \geq a$.

1.1.2 (b) If $a \geq b$ and $b \geq c$, then $a \geq c$.

Proof. Let m and n be natural numbers such that $a = b + m$ and $b = c + n$. Then we have $a = (c + n) + m$. By the associative law for addition, $a = c + (n + m)$. Therefore $a \geq c$.

1.1.3 (c) If $a \geq b$ and $b \geq a$, then $a = b$.

Proof. Let m and n be natural numbers such that $a = b + m$ and $b = a + n$. Then we have $a = (a + n) + m$ and $a = a + (n + m)$, and so $n + m$ must be zero. Thus $n = m = 0$ by Corollary 2.2.9. Substituting 0 for m into $a = b + m$, we conclude that $a = b$.

1. Lemma. If $a + m = a$ then $m = 0$.

Proof. $a + m = a = a + 0$. Using the cancellation law gives $m = 0$.

1.1.4 (d) $a \geq b$ iff $a + c \geq b + c$.

Proof.

1. (i) If $a \geq b$ then $a + c \geq b + c$.

Let $a = b + m$ for some natural number m . Then $a + c = (b + c) + m$, hence $a + c \geq b + c$.

2. (ii) If $a + c \geq b + c$ then $a \geq b$.

Since $a + c \geq b + c$, there exists a natural number m such that $a + c = (b + c) + m$. Then $a + c = (b + m) + c$. By the cancellation law $a = b + m$. Hence $a \geq b$.

1.1.5 (e) $a < b$ iff $a++ \leq b$.

Proof. First, suppose $a < b$. Then there exists a natural number m such that

$$b = a + m$$

Since $a < b$ implies $a \neq b$, $m \neq 0$. Thus by Lemma 2.2.10, there is a natural number n such that $n++ = m$. Hence

$$b = a + (n++) = (a + n)++ = (a++) + n$$

Therefore $a++ \leq b$.

Conversely, suppose that $a++ \leq b$. Let m be a natural number such that $b = (a++) + m$. Then

$$b = (a++) + m = (a + m)++ = a + (m++)$$

Suppose, for contradiction, that $a = b$. Then

$$a = a + (m++)$$

and cancelling a gives $m++ = 0$. But that contradicts Axiom 2.3, which says $m++ \neq 0$. Thus $a \neq b$.

Since $b = a + (m++)$, we conclude that $a < b$.

1.1.6 (f) $a < b$ iff $b = a + d$ for some positive number d .

Proof. First, assume that $a < b$. Then $b = a + d$ for some natural number d . Suppose, for contradiction, that $d = 0$. then $b = a + 0 = a$, contradicting the fact that $a \neq b$. Hence $d \neq 0$, and we conclude d is positive.

Conversely, assume that $b = a + d$ for some positive number d . If $a = b$, $d = 0$ by the same argument used in (e), contradicting the fact that $d \neq 0$. Thus $a \neq b$ and we conclude that $a < b$.